DRAWING LIST

ITEM	QTY	DRAWING NUMBER	DESCRIPTION	
1	1	S1101EFA2119	CAMERA STEPPER DRIVER BOARD ASSEMBLY	7
2	1	S1101EFA2120	CAMERA CONTROL POWER DISTRIBUTION BOARD ASSEMBLY	7
3	1	S1101EFA2121	CAMERA ELECTRONICS HIGH VOLTAGE POWER HARNESS ASSEMBLY	6
4	1	S1101EFA2124	CAMERA ELECTRONICS LOW VOLTAGE POWER HARNESS ASSEMBLY	6
5	1	S1101EFA2125	CAMERA ELECTRONICS MOTION CARD HARNESS I (J5) ASSEMBLY	6
6	1	S1101EFA2126	CAMERA ELECTRONICS MOTION CARD HARNESS II (J6) ASSEMBLY	6
7	1	S1101EFA2127	CAMERA ELECTRONICS MOTOR CONTROL AND COMMUNICATIONS HARNESS ASSEMBLY	6
8	1	S1101EFA2128	CAMERA ELECTRONICS FILTER MOTOR CONTROL HARNESS ASSEMBLY	6
9	1	S1101EFA2129	MODIFIED CAMERA POWER BOARD	7
10	1	S1101EFA2130	MODIFIED CAMERA MOTION CONTROL BOARD	7
11	1	S1101EFA3215	MOTION CONTROL BOARD INTERFACE ASSEMBLY	7
12	1	S1101MFA2101	CAMERA CONTROL BASE ASSEMBLY	
13	1	S1101MFA2103	CAMERA CONTROL BOARD MOUNT ASSEMBLY	
14	1	S1101MFA2105	CAMERA CONTROL LEFT SIDE ASSEMBLY	
15	1	S1101MFA2107	CAMERA CONTROL RIGHT SIDE ASSEMBLY	
16	2	S1101MFA2122	CAMERA CONTROL AIR OUTLET NUTPLATE ASSEMBLY	
17	1	S1101MFD2109	CAMERA CONTROL AIR INLET SCREEN	
18	1	S1101MFD2110	CAMERA CONTROL AIR INLET SCREEN COVER	
19	2	S1101MFD2111	CAMERA CONTROL AIR OUTLET SCREEN	
20	2	S1101MFD2112	CAMERA CONTROL AIR OUTLET SCREEN COVER	
21	1	S1101MFD2113	CAMERA CONTROL BENCH SEAL	
22	2	S1101MFD2114	CAMERA CONTROL HANDLE DRIVER PIN	
23	4	S1101MFD2115	CAMERA CONTROL HANDLE SPRING	
24	2	S1101MFD2116	CAMERA CONTROL HANDLE SPRING HOUSING	
25	1	S1101MFD2117	CAMERA CONTROL HANDLE DOCKING LATCH	
26	1	S1101MFD2118	CAMERA CONTROL COVER	

COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C1	1	CSD-1	NUTPLATE, MIL-DTL-24308 CONNECTOR, SHELL SIZE 9 .112-40 UNJC-3B THREAD, STROCO
C2	1	CSD-3	NUT PLATE, MIL-DTL-24308 CONNECTOR, SHELL SIZE 25 .112-40 UNJC-3B THREAD, STROCO
C3	2	WST-93-PA-S02	RETAINING RING, WST SERIES, 302 STAINLESS, PASSIVATED. SMALLEY
C4	4	JAX103-M09-F09-440	JACK SCREW, #4-40 X .312 LG, THREAD 303 STAINLESS STEEL, PASSIVATED, UNICORP INC.
C5	2	M21143/1-34	PIN, STRAGHT, HEADLESS, (DOWEL), CRES, TYPE 303 .125 DIA X .312 LG
C6	18	NAS620C4L	FLAT WASHER, REDUCED OUTSIDE DIAMETER NO. 4 (.1120), .016 THICK, CRES
C7	12	NAS620C6L	FLAT WASHER, REDUCED OUTSIDE DIAMETER NO. 6 (.1380), .016 THICK, CRES
C8	14	NAS1101E04-3	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .188 LENGTH
C9	2	NAS1101E04-7	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .438 LENGTH
C10	2	NAS1101E06-10	SCREW, MACHINE-FLAT FILLISTER HEAD .138-32 UNJC-3A X .625 LENGTH
C11	2	NAS1101E06-14	SCREW, MACHINE-FLAT FILLISTER HEAD .138-32 UNJC-3A X .875 LENGTH
C12	8	NAS1102E02-5	.086-56 UNJC-3A X .312", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C13	8	NAS1102E02-6	.086-56 UNJC-3A X .375", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C14	28	NAS1102E04-6	.112-40 UNJC-3A X .375", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C15	8	NAS1352N06-8	SCREW, CAP, SOCKET HEAD, A286, PASSIVATED .1380-32 UNRC-3A X .500
C16	16	NAS1829C3C10	#4-40 MALE-FEMALE STANDOFF, .188 HEX X .625 LG, 303 STAINLESS STEEL, PASSIVATED
C17	2	NAS1829C4C21	#4-40 MALE-FEMALE STANDOFF, .250 HEX X 1.313 LG, 303 STAINLESS STEEL, PASSIVATED
C18	A/R	3145 GRAY	RTV ADHESIVE/SEALANT, DOW CORNING
C19	A/R	222MS	LOCTITE 222MS THREAD LOCKER
C20	A/R	MS-907	EPOXY ADHESIVE, MILLER -STEPHENSON
C21	2	NAS1101E04-4	SCREW, MACHINE-FLAT FILLISTER HEAD .112-40 UNJC-3A X .250 LENGTH
C22	A/R	PLT1M-M702	CABLE TIE, HALAR FLUOROPOLYMER, PANDUIT CORP.
C23	A/R	TC105	CABLE TIE MOUNTING BASE, ALUMINUM, THOMAS & BETTS
C24	A/R	2216 B/A	2216 B/A GRAY EPOXY 3M SCOTCH WELD EPOXY

11. CABLE AND WIRE ROUTING AND CABLE TIE MOUNT PLACEMENT TO BE DETERMINED BY TECHNICIAN AT TIME OF ASSEMBLY.

12. ALL HARNESS CONNECTIONS TO BE STAKED USING RTV (ITEM C18) AT FINAL ASSEMBLY. STAKE ITEM 11 TO ITEM 10 USING RTV (ITEM C18).

13. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.

14. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.

15. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW PER SAE-AS478, METHOD 35D.

16. APPLY ITEM C20 PER MANUFACTURER'S INSTRUCTIONS.

17. DISCARD SUPPLIED WASHERS AND NUTS FOR JACK SCREW, ITEM C4.

18. REMOVE ALL TEMPORARY HARNESS LABELS.

19. COVER ALL SHARP EDGES AND PROTRUSIONS THAT MAY CAUSE DAMAGE TO HARNESSES USING RTV (ITEM C18).

NOTES:

1. MATERIAL AND PROCESS CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.

2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.

3. STAKING, BONDING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1

4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4.

5. ASSEMBLE AND TEST PER ACME-TST-027.

6. REFER TO THE FOLLOWING DRAWINGS FOR WIRING INTERCONNECTION:

   S1101EFA2121, CAMERA ELECTRONICS HIGH VOLTAGE POWER HARNESS ASSEMBLY
   S1101EFA2124, CAMERA ELECTRONICS LOW VOLTAGE POWER HARNESS ASSEMBLY
   S1101EFA2125, CAMERA ELECTRONICS MOTION CARD HARNESS I (J5) ASSEMBLY
   S1101EFA2126, CAMERA ELECTRONICS MOTION CARD HARNESS II (J6) ASSEMBLY
   S1101EFA2127, CAMERA ELECTRONICS MOTOR CONTROL AND COMMUNICATIONS HARNESS ASSY
   S1101EFA2128, CAMERA ELECTRONICS FILTER MOTOR CONTROL HARNESS ASSEMBLY

7. REFER TO THE FOLLOWING DRAWINGS FOR PC BOARD PINOUT REFERENCES:

   S1101EFA2119, CAMERA STEPPER DRIVER BOARD ASSEMBLY
   S1101EFA2120, CAMERA CONTROL POWER DISTRIBUTION BOARD ASSEMBLY
   S1101EFA2129, MODIFIED CAMERA POWER BOARD
   S1101EFA2130, MODIFIED CAMERA MOTION CONTROL BOARD

8. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3.

9. APPLY THREAD LOCKING COMPOUND PER ESTABLISHED PROCEDURES.

10. TORQUE VALUES NOTED ON THE DRAWING ARE EFFECTIVE TORQUE ONLY. RUNNING TORQUE IS REQUIRED FOR SELF LOCKING DEVICES AND BOLT FRICTION. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE. RECORD RUNNING TORQUE AND FINAL TORQUE ON DRAWING.

ACME-ECO-520 In Process

REV	DESCRIPTION	APPROVED
A	INCORPORATED ACME-ECO-356.	J.OGRIN 04/20/15 G.FEDEVICH 04/23/15 A.DRAKE 04/24/15

REVISION HISTORY

S1101MFA2000
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E.McCLOSKEY 04/24/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. KYNIK 07/26/13	QUALITY	D. SEELEY 07/30/13
REVIEWER	G. FEDEVICH 08/01/13	MATERIALS	M. WILKINSON 08/06/13
MECHANICAL	A. DRAKE 07/29/13	PROJ LEAD	B. BOROWSKI 08/19/13
ELECTRICAL	T. GOBELI 07/30/13		
STRUCTURAL	C. WERNER 07/31/13		
SAFETY	W. BORRELLI 07/30/13		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
8745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC09BA02B
PROJECT CODE: SPACE 110

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
CAMERA CONTROL ASSEMBLY

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA2100	A

SCALE 2/1 | INITIAL REL DATE 08/20/2013 | SHEET 1 OF 6

RELEASE LEVEL: RELEASED
MODEL: S1101MFA2100

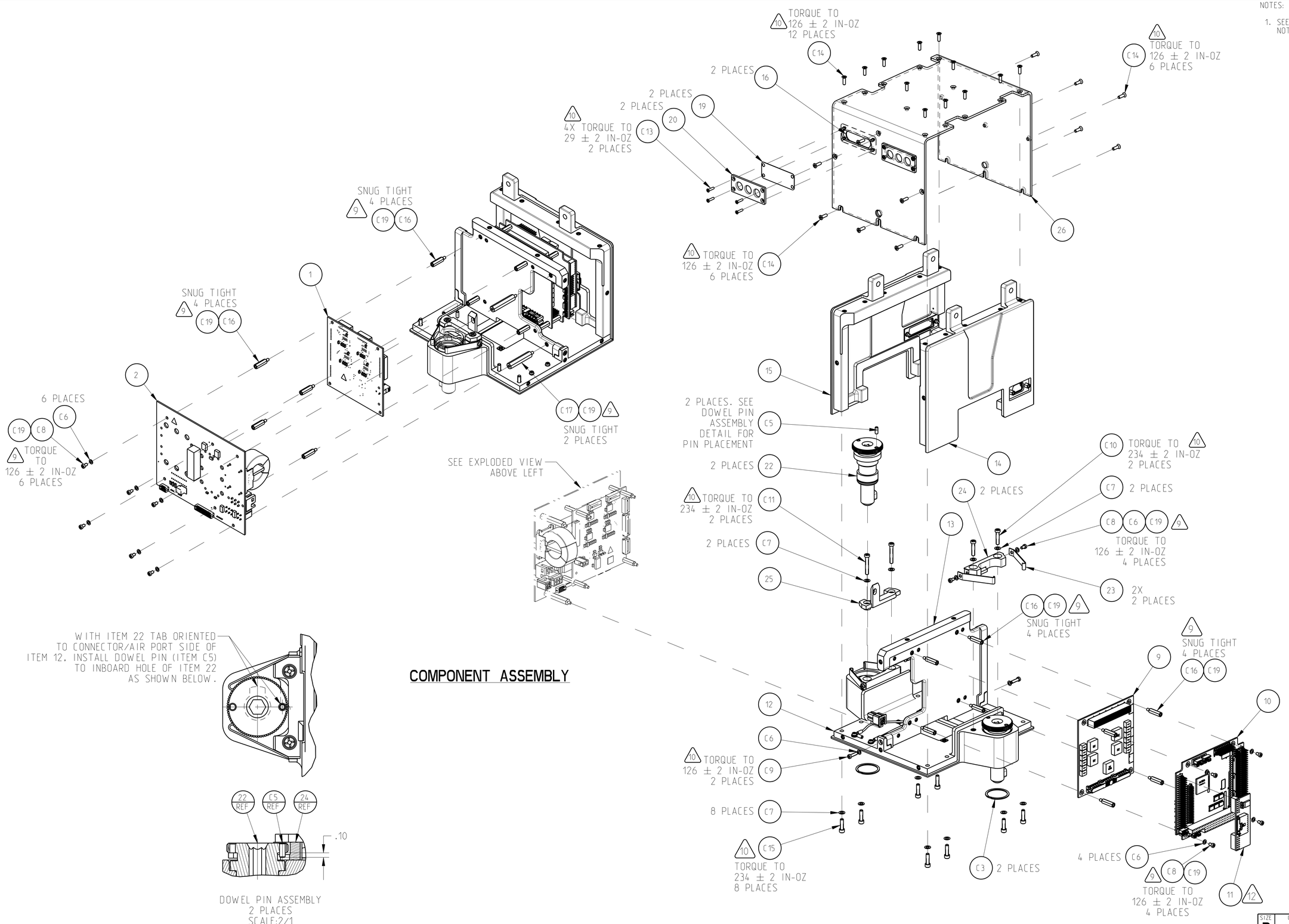
PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA2100 | SH 1 | REV A

ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.



WITH ITEM 22 TAB ORIENTED TO CONNECTOR/AIR PORT SIDE OF ITEM 12, INSTALL DOWEL PIN (ITEM C5) TO INBOARD HOLE OF ITEM 22 AS SHOWN BELOW.

DOWEL PIN ASSEMBLY
2 PLACES
SCALE:2/1

COMPONENT ASSEMBLY

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA2100	A
SCALE 1/2	INITIAL REL DATE 08/20/2013	SHEET 2 OF 6	

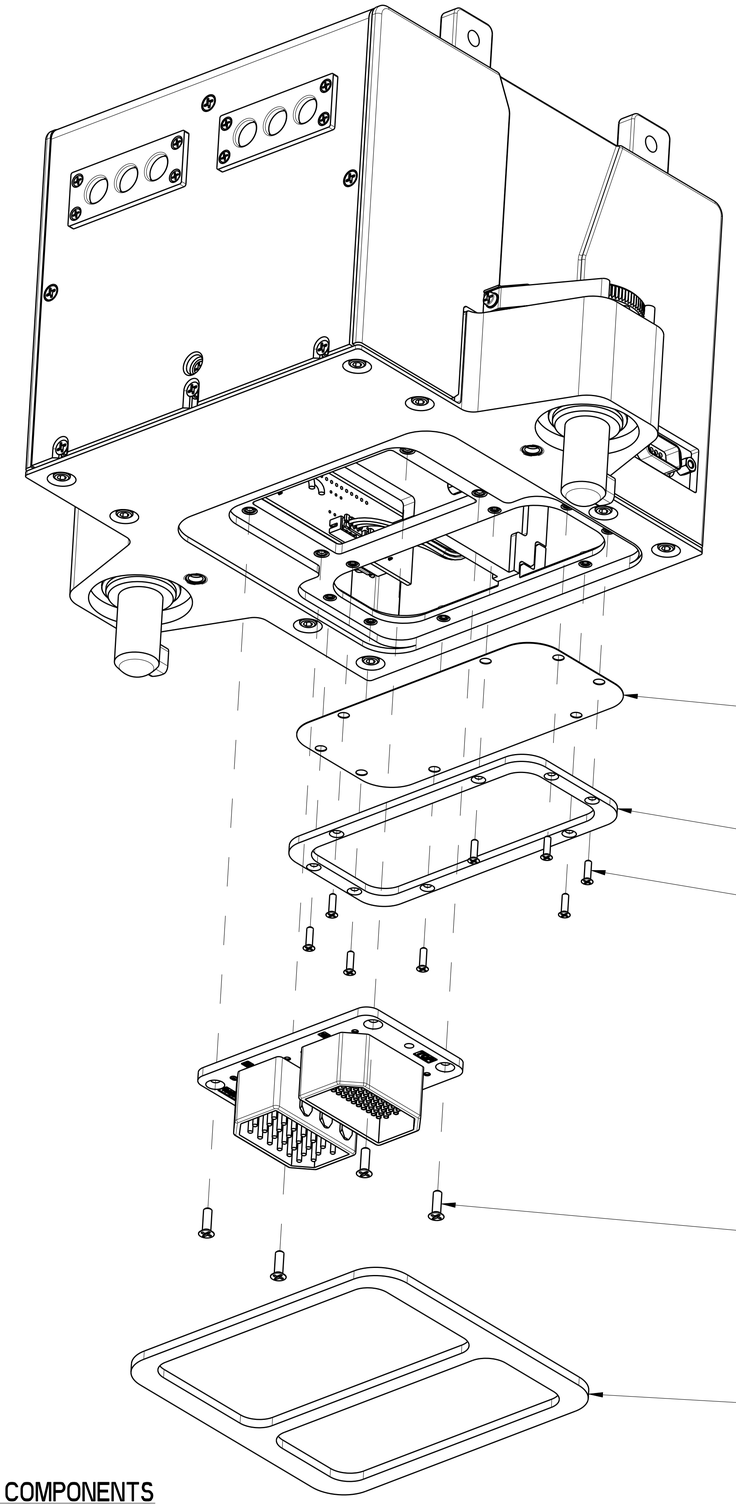
PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA2100 | SH 2 | REV A

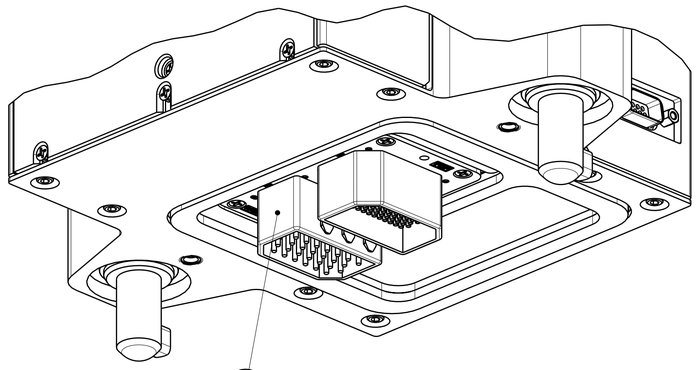
ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.

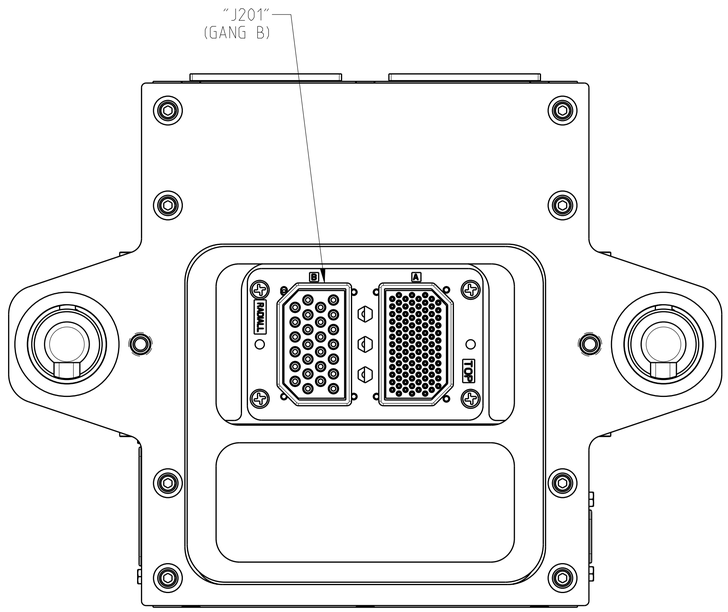


INTERFACE COMPONENTS



HIGH VOLTAGE POWER "J201"
ITEM 3 REF. "UML6"

INTERFACE CONNECTION



CONNECTOR ORIENTATION

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA2100	A
SCALE 1/1	INITIAL REL DATE 08/20/2013	SHEET 3 OF 6	

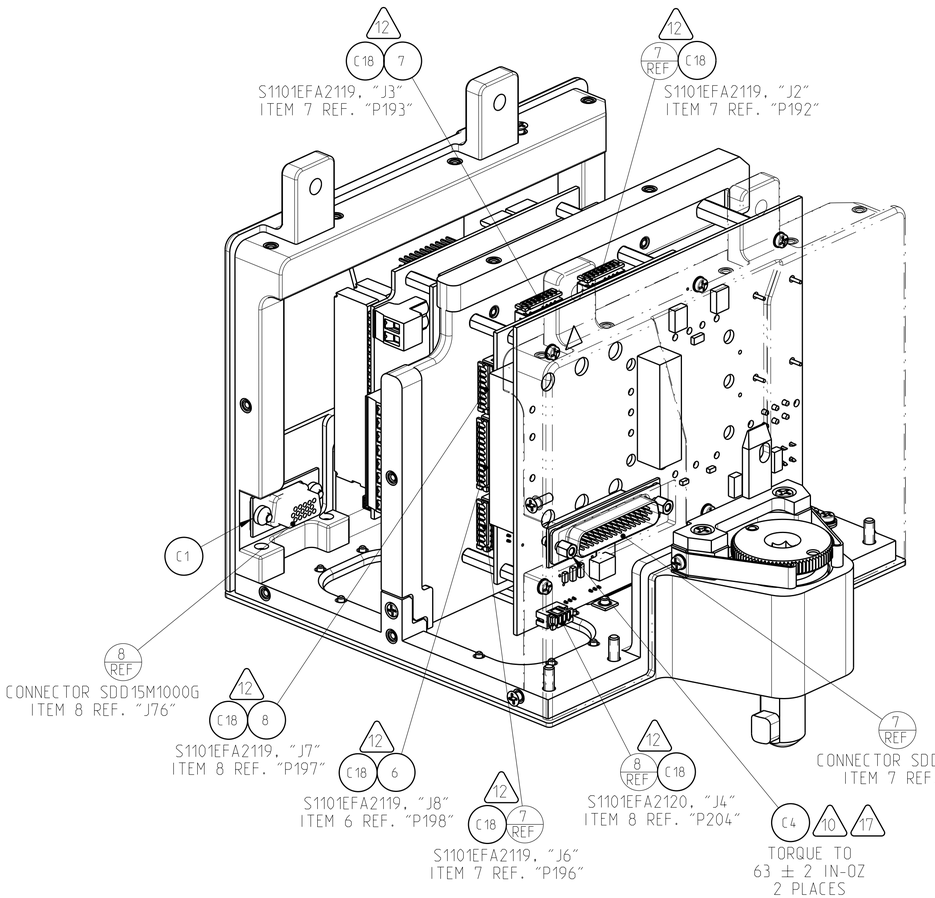
PRO/E VERSION: Wildfire 2.0

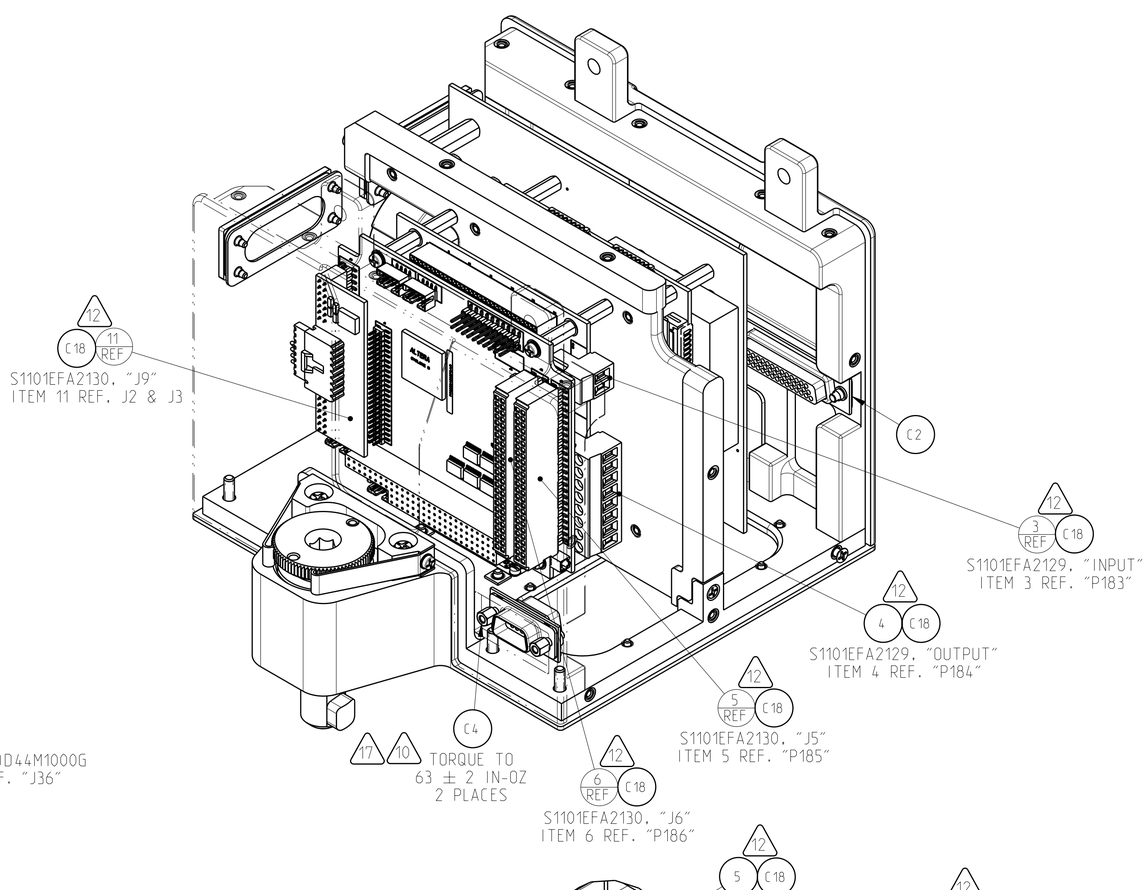
DWG NO S1101MFA2100 | SH 3 | REV A

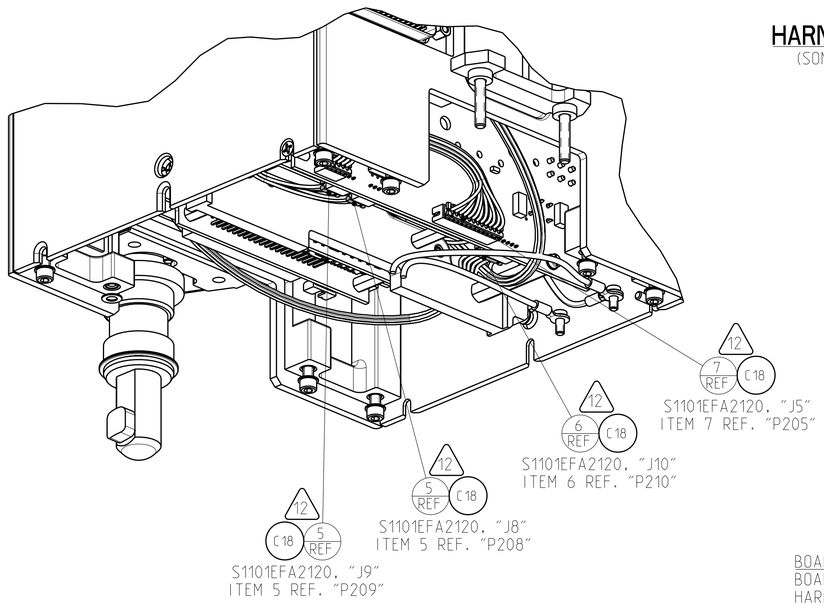
ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.

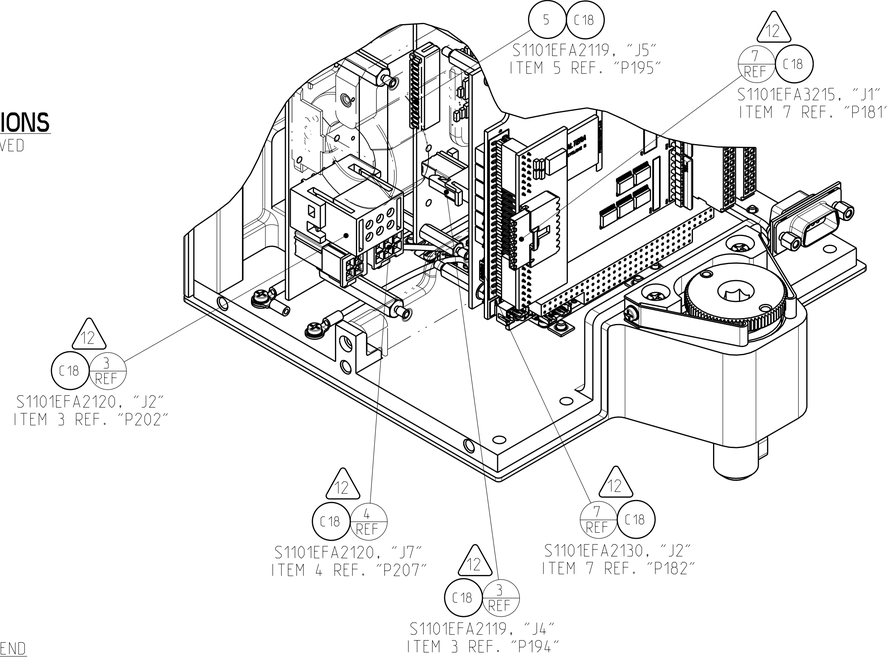






HARNESS CONNECTIONS

(SOME COMPONENTS REMOVED FOR CLARITY)



BOARD AND HARNESS LEGEND
BOARD DRAWING NUMBER. "CONNECTOR LOCATION NUMBER"
HARNESS DRAWING NUMBER. "CONNECTOR NUMBER"

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA2100	A
SCALE 1/1		INITIAL REL DATE 08/20/2013 SHEET 4 OF 6	

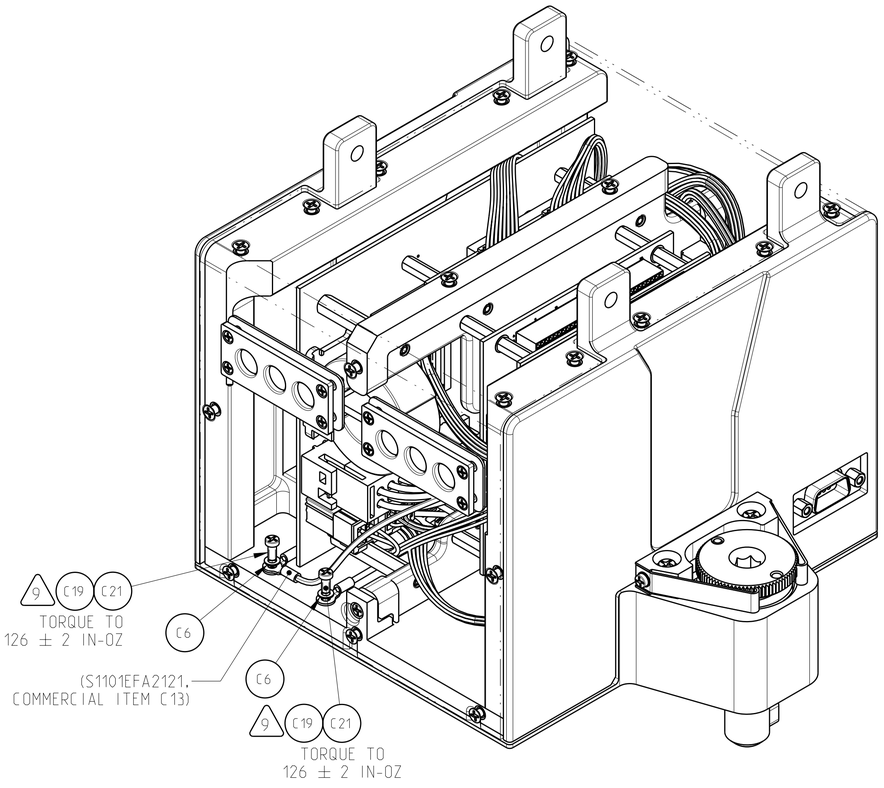
PRO/E VERSION: Wildfire 2.0

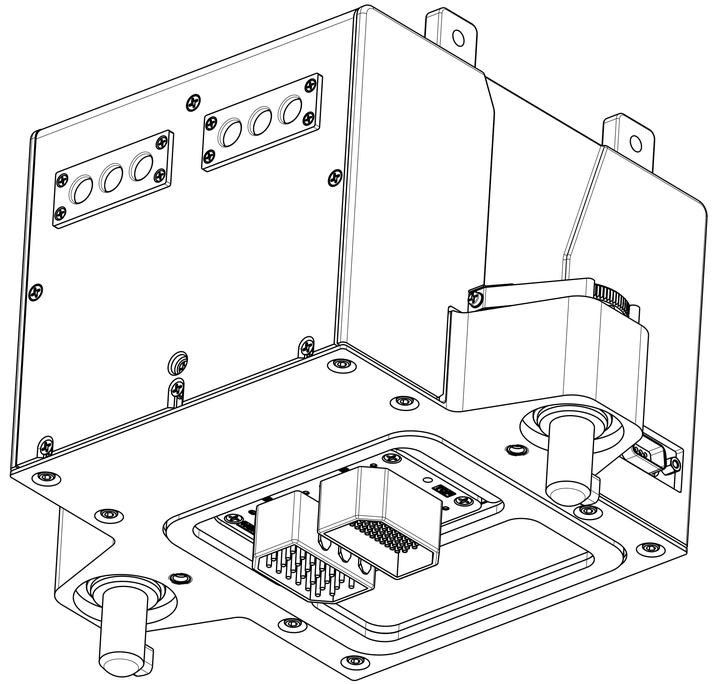
DWG NO S1101MFA2100 SH 4 REV A

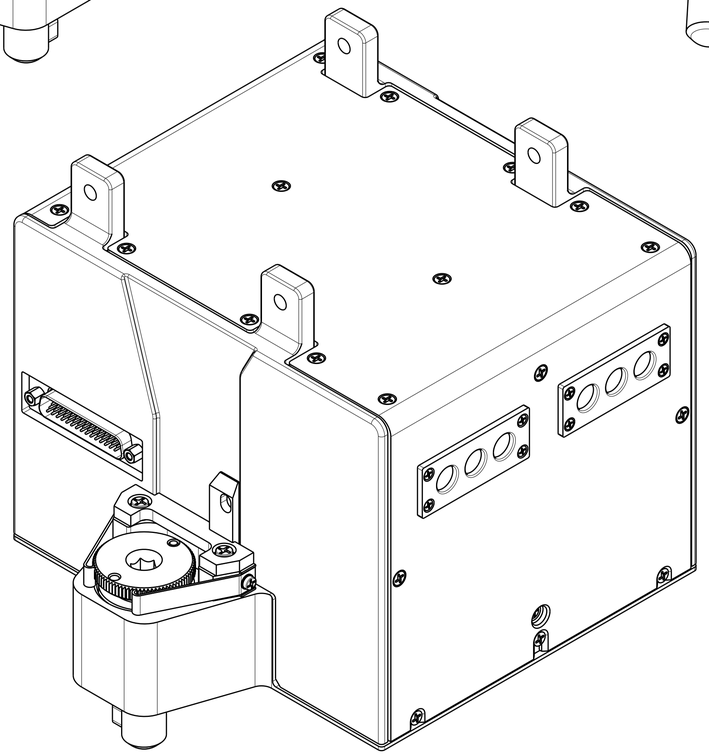
ZIN_D_2012.frm

NOTES:

1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.







SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA2100	A
SCALE 1/1	INITIAL REL DATE 08/20/2013	SHEET 5 OF 6	

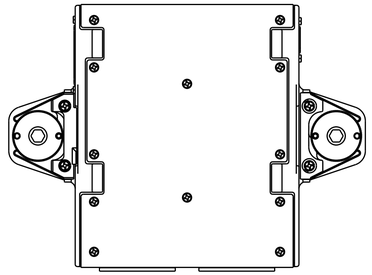
PRO/E VERSION: Wildfire 2.0

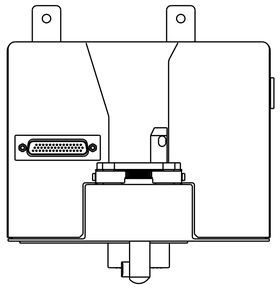
DWG NO S1101MFA2100 SH 5 REV A

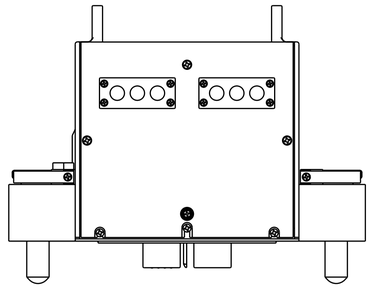
ZIN_D_2012.frm

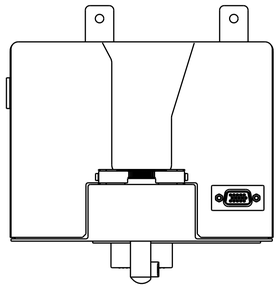
NOTES:

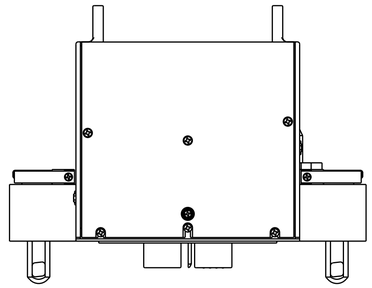
1. SEE SHEET 1 FOR ADDITIONAL INFORMATION. GENERAL NOTES AND TOLERANCES.

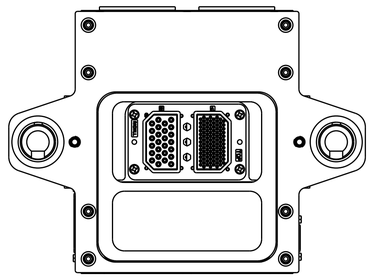












SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101MFA2100	A
SCALE 1/2	INITIAL REL DATE 08/20/2013	SHEET 6 OF 6	

PRO/E VERSION: Wildfire 2.0

DWG NO S1101MFA2100 | SH 6 | REV A

ZIN_D_2012.frm